

Problem 3

To pull a car out of the mud, the driver uses a rope, tying one end to the front of the car and the other end to a nearby tree, which is at a distance of $d = 12$ m from the car. The driver, with a mass of $m = 80$ kg, hangs on the middle of the rope, causing a sag of $\Delta h = 0.25$ m. If the rope was initially horizontal and aligned with the car's axis, determine the force exerted on the car in this situation.

Solution

The driver exerts a downward force due to gravity:

$$F_g = mg = 80 \times 9.81 = 784.8 \text{ N}$$

Since the driver hangs in the middle, the rope forms two symmetrical right triangles. The horizontal distance to the midpoint is half the total distance:

$$L = \frac{d}{2} = \frac{12}{2} = 6 \text{ m}$$

The rope's vertical displacement forms the opposite side of the triangle, so we calculate the rope's length on each side:

$$\begin{aligned}\ell &= \sqrt{L^2 + (\Delta h)^2} = \sqrt{6^2 + (0.25)^2} \\ \ell &= \sqrt{36 + 0.0625} = \sqrt{36.0625} \approx 6.01 \text{ m}\end{aligned}$$

The angle the rope makes with the horizontal is:

$$\theta = \tan^{-1} \left(\frac{\Delta h}{L} \right) = \tan^{-1} \left(\frac{0.25}{6} \right)$$

Since θ is small, we can approximate the tension components. The vertical equilibrium condition tells us:

$$2T \sin(\theta) = mg$$

Using the small-angle approximation ($\sin(\theta) \approx \frac{\Delta h}{\ell}$):

$$2T \frac{0.25}{6.01} = 784.8$$

Solving for T :

$$\begin{aligned}T &= \frac{784.8 \times 6.01}{2 \times 0.25} \\ T &= \frac{4714.45}{0.5} = 9428.9 \text{ N}\end{aligned}$$

The horizontal force on the car is the horizontal component of the tension:

$$F_{\text{car}} = 2T \cos(\theta)$$

Since $\cos(\theta) \approx 1$ for small angles:

$$F_{\text{car}} \approx 2 \times 9428.9 = 18857.8 \text{ N}$$

Result: The force exerted on the car is approximately:

$$F_{\text{car}} \approx 18858 \text{ N}$$